FinPlan-LLM

A Multi-Agent Financial Advisory System

DS 5983: Introduction to Large Language Models

Khoury College of Computer Sciences

Harsh Vijay Mamania

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LangGraph | Gemini 2.5 Flash | LangSmith

Motivation: Baseline (Pure LLM response)

Here is a portfolio allocation tailored to your client's profile:

1. **Asset Allocation Percentages:**
   * **Stocks:** 15%
   * **Bonds:** 65%
   * **Cash Equivalents:** 20%

2. **Specific ETF Recommendations:**
   * **Stocks (15%):**
     * **VOO** (Vanguard S&P 500 ETF)
   * **Bonds (65%):**
     * **BND** (Vanguard Total Bond Market ETF) - 45%
     * **VGSH** (Vanguard Short-Term Government Bond ETF) - 20%
   * **Cash Equivalents (20%):**
     * **SGOV** (iShares 0-3 Month Treasury Bond ETF)

3. **Brief Rationale:**
   This allocation prioritizes capital preservation and stability, aligning with the client's conservative risk tolerance and 7-year time horizon. The low stock allocation (VOO) provides diversified exposure to large-cap U.S. companies for modest growth potential. The dominant bond allocation (BND for broad investment-grade bonds and VGSH for ultra-safe short-term government bonds) aims to generate income and significantly reduce volatility. The substantial cash equivalent position (SGOV) offers maximum liquidity and absolute capital preservation, crucial for a client nearing retirement with a priority on protecting their principal.

Motivation: Why Agentic Financial Planning?

The Problem: Naive LLM Output

Prompt: "Create portfolio for 62yo conservative investor, 7yr horizon"

Stocks: 15% (VOO)
Bonds: 65% (BND 45%, VGSH 20%)
Cash: 20% (SGOV)

"This allocation prioritizes capital preservation..."

Issues with This Approach

1. No real market data (uses training knowledge)
2. No validation or backtesting
3. Cannot handle dynamic constraints
4. No explainability or reasoning trace
5. Hallucination risk in financial advice

Our Solution: Agentic Workflow

Build a multi-agent system that autonomously:

- Parses natural language constraints
- Fetches real-time market data (yFinance, FRED)
- Researches and compares ETFs autonomously
- Validates decisions with self-reflection
- Explains every step of reasoning

Key Design Principle

"LLMs for reasoning and soft decisions, Heuristics for hard financial decisions"

Best of both worlds: Intelligence + Reliability

"Build an agentic LLM that autonomously selects and uses external tools"

System Architecture

USER INPUT
age, risk, goal
horizon, constraints

→

PHASE 1: UNDERSTANDING

Constraint Parser
LLM: NLU extraction
→ exclude_tags, require_esg
→ filtered_universe (41 ETFs)

Macro Analysis
API: FRED (fed_rate, CPI...)
LLM: Regime classification
→ macro_regime

↓ parsed_constraints, macro_data, filtered_universe

PHASE 2: RESEARCH AGENT (ReAct Pattern)

CORE AGENTIC

LLM Agent
Autonomous tool selection
Dynamic sequencing
recursion_limit: 50

↔

screen_asset	compare_etfs	get_details	fetch_news
Filter by class	Metrics comparison	ETF metadata	Yahoo RSS

→ top_picks: {US_Stocks: VTI, Bonds: BND, International: VEA, REITs: VNQ, Commodities: GLD}

↓ top_picks, research_output

PHASE 3: DECISION PIPELINE

Allocator
Heuristics: 100-age rule
+ regime adjustments
→ final_allocation %

→

Backtester
API: yFinance historical
Sharpe, MaxDD, Vol
→ backtest_metrics

→

News Analyzer
LLM: Summarization
Sentiment extraction
→ news_sentiment

↓ final_allocation, backtest_metrics, news_analysis

PHASE 4: REFLECTION LOOP (max 2 iterations)

Validator (LLM-as-Judge)
Chain-of-Thought validation
Age, Risk, Goal, Constraints

should_continue?
concerns AND attempts < 2

Rebalancer
LLM: Diagnose issue
→ preset_override
↩ back to Allocator

OUTPUT

PDF Report
Allocation, Backtest, Reasoning

Console Traces
Step-by-step execution logs

LangSmith
Full LLM call traces

FinancialPlanState

user_profile: dict
parsed_constraints: dict
excluded_etfs: list
macro_data: dict
top_picks: dict
final_allocation: dict
backtest_metrics: dict
validation_result: str
rebalance_attempts: int

Tech Stack

LLM: Gemini 2.5 Flash
Orchestration: LangGraph
Data: yFinance, FRED
Observability: LangSmith

Legend
NLU · Agentic · Heuristics · Reflection

Phase 1: Understanding

Constraint Parser Node

LLM Concept: NLU / Structured Extraction

Input (Natural Language):

"No oil companies, ESG focus, avoid China exposure"

LLM Processing:

1. Parse constraint text with structured prompt
2. Map to CONSTRAINT_KEYWORDS dictionary
3. Extract: exclude_tags, require_esg, preferences
4. Handle unmatched constraints gracefully

Output (Structured JSON):

exclude_tags: ["oil", "fossil_fuel", "china_exposure"]
require_esg: true
excluded_etfs: [XLE, VDE, EEM, VWO...] (8 total)
filtered_universe: 41 ETFs remaining

ETF_METADATA System

49 ETFs with tags: oil, fossil_fuel, tobacco, weapons, china_exposure, esg, emerging_markets, etc.

Macro Analysis Node

LLM Concept: Summarization / Classification

Data Source: FRED API

Fed Rate: 4.33%
Inflation (CPI): 3.2%
Unemployment: 4.4%
GDP Growth: 3.8%

LLM Interpretation:

1. Analyze all indicators holistically
2. Classify economic regime
3. Generate investment implications
4. Identify key risks

Output: macro_regime

Possible regimes:

Goldilocks | Stagflation | Recession | Strong Growth | High Inflation | Rising Rates

State Passed to Next Phase:

parsed_constraints, excluded_etfs
filtered_universe, macro_data, macro_regime

Phase 2: Research Agent

CORE AGENTIC

ReAct Pattern
LLM autonomously selects tools

1. OBSERVE
user_profile
excluded_etfs
prefer_esg
filtered_universe

2. THINK
"Screen US_Stocks first"
"Compare top 2 candidates"
"Check news for risk"

3. ACT
screen_asset_class(
asset_class,
excluded, prefer_esg)

↻ Repeat until all classes done

Output: top_picks
US_Stocks: VTI
International: VEA
Bonds: BND
REITs: VNQ
Commodities: GLD

Available Tools (4) - Internal Logic

screen_asset_class(asset_class, excluded_etfs, prefer_esg)

Process:
1. Filter ETF_METADATA by asset_class
2. Remove tickers in excluded_etfs list
3. Fetch 6mo metrics via yFinance
4. Rank by: YTD return, Sharpe ratio, expense
5. If prefer_esg: boost ESG-tagged ETFs

Returns (top 3):
ticker, YTD%, Sharpe, expense_ratio
volatility, max_drawdown, ESG_score

compare_etfs(ticker1, ticker2)

Comparison Metrics:
YTD return, 1Y return, 3Y return
Sharpe ratio, Sortino ratio
Max drawdown, Volatility (std dev)
Expense ratio, AUM, Dividend yield

Returns:
Side-by-side table
Winner per metric highlighted
Overall recommendation with reason

get_etf_details(ticker)

Data Retrieved:
Full name, Category, Index tracked
Top 10 holdings with weights
Sector breakdown percentages

Additional Info:
Geographic exposure
ESG rating (if available)
Tags from ETF_METADATA

fetch_market_news(query)

Source:
Yahoo Finance RSS feed
Query: ticker or topic string
Limit: 5 most recent articles

Returns per article:
Headline, Summary snippet
Publication timestamp
Source URL for citation

LLM Concepts: Autonomous Tool Selection, ReAct Pattern, Multi-step Reasoning, Dynamic Sequencing

Phase 3: Decision Pipeline

Allocator Node

Deterministic Heuristics

Base Allocation:

100 - age = equity %
Risk tolerance adjustment
Goal-based modifications

Regime Adjustments:

Stagflation: +15% Commodities
Recession: +15% Bonds
Strong Growth: +10% Stocks

Output:

final_allocation: dict
{US_Stocks: 38.5%,
Bonds: 25.5%, ...}

Why Heuristics? Auditable, deterministic, no hallucination

Backtester Node

API: yFinance Historical

Process:

1. Fetch historical prices (1Y, 3Y)
2. Calculate weighted portfolio returns
3. Compare vs S&P 500 benchmark

Metrics Calculated:

Total Return, Annualized Return
Sharpe Ratio (risk-adjusted)
Max Drawdown, Volatility
Best Day, Worst Day

Sample Output:

1Y Return: +16.44%
Sharpe: 1.16
vs SPY: +2.25%

News Analyzer Node

LLM: Summarization + Sentiment

Process:

1. Fetch news for top 4 holdings
2. LLM summarizes key developments
3. Extract sentiment + confidence

LLM Output Fields:

NEWS_SUMMARY (3-4 sentences)
SENTIMENT: BULLISH/BEARISH/NEUTRAL
CONFIDENCE: HIGH/MEDIUM/LOW
PORTFOLIO_IMPACT
KEY_RISKS, SOURCES

LLM Concepts:

Summarization, Sentiment Analysis
Grounding/Citation

State Output: final_allocation, backtest_metrics (1Y, 3Y), news_analysis, news_sentiment

Allocation Heuristics: Deep Dive

Step 1: Base Allocation (100 - Age)

Age 30 = 70% equity, 30% bonds
Age 50 = 50% equity, 50% bonds
Age 65 = 35% equity, 65% bonds

↓

Step 2: Risk Tolerance Adjustment

Conservative: -20% equity, +20% bonds
Moderate: No change (baseline)
Aggressive: +20% equity, -20% bonds

↓

Step 3: Detect Macro Regime

Based on GDP, Inflation, Unemployment
Classify: Stagflation, Recession, Growth...

↓

Step 4: Apply Regime Shifts + Normalize

Apply adjustments from matrix
Ensure non-negative, normalize to 100%

Regime Adjustment Rules

Stagflation: -25% Stocks, +10% Bonds, +15% Commodities

Recession: -20% Stocks, +15% Bonds, +5% Commodities

Strong Growth: +10% Stocks, -10% Bonds, -5% Commodities

High Inflation: -15% Stocks, +10% Bonds, +10% Commodities

Goldilocks: No adjustments (ideal conditions)

Example: 30yo, Moderate, Strong Growth

Base (age 30): 70% equity
Risk adjustment: +0% (moderate)
Regime shift: +10% stocks

Final Allocation:

US Stocks: 38.5% (VTI) | Bonds: 25.5% (BND)
International: 17.5% (VEA) | REITs: 10.5% (VNQ)
Commodities: 8.0% (GLD)

Why Heuristics over LLM?

Deterministic, auditable, no hallucination risk in financial decisions

Phase 4: Reflection Loop

Validator Node (LLM-as-Judge)

LLM Concept: Chain-of-Thought Validation

5-Step Validation Process:

Step 1: Age Appropriateness
Does equity % match 100-age guideline?

Step 2: Goal Alignment
Does structure support stated goal + horizon?

Step 3: Risk Tolerance Match
Is volatility/drawdown appropriate for risk level?

Step 4: Constraint Compliance
Are all user constraints satisfied?

Step 5: Risk Metrics
Are Sharpe, MaxDD, Vol acceptable?

Output: APPROVED or CONCERNS NOTED

should_continue?

IF concerns exist
AND attempts < 2

YES ↓

Rebalance

Reasoning
+ Report

NO →

Rebalancer Node

LLM Concept: Self-Correction

LLM Diagnosis:

Analyzes validation feedback and classifies issue:

TOO_AGGRESSIVE
Shift to conservative preset

TOO_CONSERVATIVE
Shift to aggressive preset

CONCENTRATION_RISK
Use moderate preset

CONSTRAINT_VIOLATION
Flag for manual review

After fix: Loop back to Allocator, re-run backtest, re-validate

Escape Hatch: Max 2 attempts
Prevents infinite loops, adds disclaimer if unresolved

LLM Concepts Demonstrated

Core Agentic Patterns

Autonomous Tool Selection
Research Agent decides which tools to call
Location: Phase 2

ReAct Pattern
Reasoning + Acting loop
Location: Research Agent

Multi-step Reasoning
Agent sequences multiple tool calls
Location: Research Agent

Multi-Agent Orchestration
8 nodes, coordinated via LangGraph
Location: Full workflow

NLU + Analysis

Structured Extraction
Natural language to JSON constraints
Location: Constraint Parser

Summarization
Condense news, macro data
Location: Macro, News Analyzer

Sentiment Analysis
BULLISH/BEARISH/NEUTRAL + confidence
Location: News Analyzer

Grounding / Citation
News sources cited in output
Location: News Analyzer

Validation + Reflection

LLM-as-Judge
Evaluates portfolio quality
Location: Validator

Chain-of-Thought
Step-by-step validation reasoning
Location: Validator

Self-Correction
Diagnose and fix issues
Location: Rebalancer

Conditional Routing
Dynamic graph edges based on output
Location: should_continue

11 LLM concepts demonstrated across 8 nodes with genuine autonomous behavior

Explainability: Three Layers

Why Explainability Matters for Agentic Workflows

Financial decisions have real consequences. Users need to trust and understand recommendations. Developers need to debug and improve.

Layer 1: Console Logs

Audience: Developer (Debugging)

```
Step: Parsing Constraints
Parsed: {exclude: [oil]}
Excluded: XLE, VDE...
Step: Research Agent
Tool: screen_asset_class
Tool: compare_etfs
Final: {VTI, BND...}
Step: Validation
Status: APPROVED
```

Real-time execution visibility

Print statements at each node

Layer 2: PDF Report

Audience: End User

Report Sections:

1. Executive Summary
2. Portfolio Allocation + Pie Chart
3. Constraint Analysis
4. Economic Analysis
5. Research Summary
6. News + Sentiment
7. Backtest + Performance Chart
8. Decision Reasoning
9. Validation Results

Professional, downloadable

Generated with ReportLab

Layer 3: LangSmith

Audience: Developer (Observability)

Trace Features:

- Waterfall view of all nodes
- Full LLM prompts + responses
- Tool call inputs/outputs
- Latency per step
- Token counts + cost
- Nested agent traces
- Shareable public links

Full observability platform

LangChainTracer integration

Demo will showcase all three layers: Console output, PDF report download, LangSmith trace walkthrough

Conclusion and Future Directions

What We Built

Multi-agent financial advisory system with genuine agentic capabilities

Key Achievements:

Autonomous ETF research via tool selection
Real-time market data integration
Natural language constraint parsing
Self-validating with reflection loop
Full explainability at every step

Tech Stack:
LangGraph, Gemini 2.5 Flash, yFinance, FRED, LangSmith, Gradio

Future Directions

More Data Sources
Technical indicators, earnings data, SEC filings, alternative data

Enhanced Tools
Portfolio optimization, risk simulation, tax-loss harvesting

User Feedback Loop
Learn from corrections, personalize over time

Multi-turn Conversation
Iteratively refine plan through dialogue

Broader Asset Classes
Crypto, individual stocks, options strategies

Key Design Insight

"LLMs for reasoning and soft decisions, Heuristics for hard financial decisions"

Best of both: Intelligence + Reliability

"Build an agentic LLM that autonomously selects and uses external tools"

11 LLM concepts, 8 nodes, 4 tools, conditional routing

Thank You! Questions?